

CSA1712

ARTIFICIAL INTELLIGENCE

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CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

(a) Propositional Logic Representation

Symbols Used

- **F** = Patient has Fever
- **C** = Patient has Cough
- **B** = Patient has Breathlessness
- **R** = Patient has Rash
- **Flu** = Patient has Flu
- **Measles** = Patient has Measles
- **Pneumonia** = Patient has Pneumonia

Rules

1. If Fever and Cough, then Flu
 $F \wedge C \rightarrow \text{Flu}$
2. If Fever and Rash, then Measles
 $F \wedge R \rightarrow \text{Measles}$
3. If Cough and Breathlessness, then Pneumonia
 $C \wedge B \rightarrow \text{Pneumonia}$

Facts for Patient A

- **F = True**
- **C = True**
- **B = True**

Therefore, the Knowledge Base is:

KB = {F, C, B, $F \wedge C \rightarrow \text{Flu}$, $F \wedge R \rightarrow \text{Measles}$, $C \wedge B \rightarrow \text{Pneumonia}$ }

(b) Forward Chaining

Initial Facts

{F, C, B}

Step 1

Rule fired: $F \wedge C \rightarrow \text{Flu}$

Since F and C are true, derive:

Flu

Updated facts:

{F, C, B, Flu}

Step 2

Rule checked: $F \wedge R \rightarrow \text{Measles}$

Rash is not available; hence no inference.

Step 3

Rule fired: $C \wedge B \rightarrow \text{Pneumonia}$

Since C and B are true, derive:

Pneumonia

Updated facts:

{F, C, B, Flu, Pneumonia}

Final Diagnoses

- Flu
- Pneumonia

Measles cannot be concluded.

(c) Backward Chaining for Goal Pneumonia

Goal: **Pneumonia**

Using rule:

$C \wedge B \rightarrow \text{Pneumonia}$

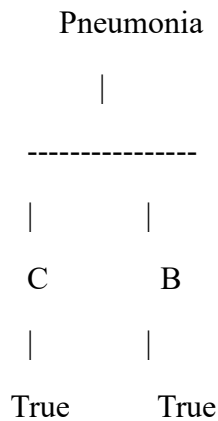
Required subgoals:

- C

- B

Both C and B are already true facts.

Goal Reduction Tree



Conclusion

Patient A has Pneumonia according to the knowledge base.

CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

(a) Unification

Rule 1

$\forall x [\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)]$

Facts:

- $\text{Vehicle}(\text{Ambulance})$
- $\text{EmergencyType}(\text{Ambulance})$

Step 1

Match $\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance})$

Substitution:

$\theta_1 = \{x/\text{Ambulance}\}$

Step 2

Apply θ_1 to $\text{EmergencyType}(x)$:

$\text{EmergencyType}(\text{Ambulance})$

Matches the fact.

Most General Unifier

$\theta = \{x/\text{Ambulance}\}$

Derived fact:

ClearPath(Ambulance)

(b) Resolution Proof for Proceed(CarA)

CNF Conversion

Rule 1:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule 2:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

Rule 3:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts:

- $\text{Vehicle}(\text{Ambulance})$
- $\text{EmergencyType}(\text{Ambulance})$
- $\text{Vehicle}(\text{CarA})$
- $\text{Behind}(\text{CarA}, \text{Ambulance})$

Negated goal:

$\neg \text{Proceed}(\text{CarA})$

Resolution Steps

1. From Rule 1 and facts derive **ClearPath(Ambulance)**.
2. From Rule 2 derive **GreenSignal(Ambulance)**.
3. Apply Rule 3 with $x=\text{CarA}$ and $y=\text{Ambulance}$ and resolve with facts to derive **Proceed(CarA)**.
4. Resolve $\text{Proceed}(\text{CarA})$ with $\neg \text{Proceed}(\text{CarA})$ to obtain $\square$.

Conclusion

$\text{Proceed}(\text{CarA})$ is proved true.

(c) Analysis for Two Emergency Vehicles

The current KB can infer GreenSignal for multiple emergency vehicles independently but cannot resolve conflicts when two emergency vehicles arrive simultaneously.

Additional Rule

$\forall x \forall y [EmergencyType(x) \wedge EmergencyType(y) \wedge Priority(x) > Priority(y) \rightarrow GreenSignal(x) \wedge Wait(y)]$

Additional Facts

- $Priority(Ambulance1)=2$
- $Priority(FireTruck1)=1$

Justification

Priority rules prevent contradictory green signals and ensure safe emergency routing.

CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

(a) CNF Conversion

P1

$SoilDry \rightarrow IrrigationNeeded$

Eliminate implication:

$\neg SoilDry \vee IrrigationNeeded$

P2

$IrrigationNeeded \wedge CropWheat \rightarrow ApplyDripMethod$

Eliminate implication:

$\neg (IrrigationNeeded \wedge CropWheat) \vee ApplyDripMethod$

Apply De Morgan's law:

$\neg IrrigationNeeded \vee \neg CropWheat \vee ApplyDripMethod$

P3

$\neg ApplyDripMethod \wedge CropWheat \rightarrow CropAtRisk$

Eliminate implication:

$\neg (\neg ApplyDripMethod \wedge CropWheat) \vee CropAtRisk$

Apply De Morgan's law:

$ApplyDripMethod \vee \neg CropWheat \vee CropAtRisk$

Facts

- SoilDry
- CropWheat

Final CNF Set

1. $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$
2. $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$
3. $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$
4. SoilDry
5. CropWheat

(b) Resolution Refutation for ApplyDripMethod

Negated goal:

$\neg \text{ApplyDripMethod}$

Resolution

1. Resolve ($\neg \text{SoilDry} \vee \text{IrrigationNeeded}$) with SoilDry $\rightarrow$ **IrrigationNeeded**
2. Resolve ($\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$) with IrrigationNeeded $\rightarrow$ **$\neg \text{CropWheat} \vee \text{ApplyDripMethod}$**
3. Resolve with CropWheat $\rightarrow$ **ApplyDripMethod**
4. Resolve with $\neg \text{ApplyDripMethod} \rightarrow \square$

Resolution Tree

SoilDry

|

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

|

IrrigationNeeded

|

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

|

$\neg \text{CropWheat} \vee \text{ApplyDripMethod}$

|

CropWheat

|

ApplyDripMethod

|

$\neg$ ApplyDripMethod

|

□

Conclusion

ApplyDripMethod is proved.

(c) Removing SoilDry

Remaining facts:

- CropWheat

Without SoilDry, IrrigationNeeded cannot be derived. Therefore ApplyDripMethod cannot be derived. Since P3 requires $\neg$ ApplyDripMethod as a fact, CropAtRisk also cannot be concluded.

Conclusion

After removing SoilDry, neither ApplyDripMethod nor CropAtRisk follows from the KB.

CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

(a) Belief State and Modus Ponens

Percepts

- Stench(1,2)
- Breeze(1,1)
- Glitter(2,2)

Rules

1. Stench(1,2) $\rightarrow$ Wumpus adjacent to (1,2)
2. Breeze(1,1) $\rightarrow$ Pit adjacent to (1,1)
3. Glitter(x,y) $\rightarrow$ Gold(x,y)

Derived Facts

- Wumpus adjacent to (1,2)
- Pit adjacent to (1,1)
- Gold at (2,2)

(b) Forward Chaining

From Stench(1,2)

Adjacent cells: (1,1), (1,3), (2,2)

Possible Wumpus locations:

- W(1,1)
- W(1,3)
- W(2,2)

From Breeze(1,1)

Adjacent cells: (1,2), (2,1)

Possible Pit locations:

- P(1,2)
- P(2,1)

Final Possible Locations

Wumpus: {(1,1), (1,3), (2,2)}

Pit: {(1,2), (2,1)}

(c) Safety of Path [1,1] → [1,2] → [2,2]

- Cell (1,2) is a possible pit.
- Cell (2,2) is a possible Wumpus location.
- Gold is present at (2,2), but danger is not ruled out.

Conclusion

The path cannot be guaranteed safe. The agent should not move to (1,2) or (2,2) without additional percepts or reasoning.